

DevOps Monitoring Setup Guide

Prometheus + Grafana on AWS EC2

Step-by-Step Installation, Integration & Dashboard Guide

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EC2 IP	100.53.31.202
Jenkins	http://100.53.31.202:8080/
Web App	http://100.53.31.202:3000/
Prometheus	http://100.53.31.202:9090/
Grafana	http://100.53.31.202:3030/

ARCHITECTURE DECISION: Both Grafana and Prometheus will be installed on your EXISTING EC2 instance (same server as Jenkins). No new instance is needed for a basic monitoring setup. Grafana will run on port 3030 (since 3000 is used by your web app).

Architecture Overview

Before starting, understand what you are building:



How it works:

- Node Exporter runs on your EC2 and exposes system metrics (CPU, RAM, Disk) on port 9100
- Prometheus scrapes (collects) those metrics every 15 seconds and stores them
- Grafana connects to Prometheus as a data source and displays beautiful dashboards
- Jenkins metrics can also be scraped by Prometheus using a plugin

Pre-Requisites: Open EC2 Security Group Ports

BEFORE anything else, you MUST open these ports in your AWS EC2 Security Group. Without this, you cannot access the tools from your browser.

Go to AWS Console > EC2 > Your Instance > Security > Security Groups > Edit Inbound Rules

Port	Protocol	Source	Purpose
9090	TCP	0.0.0.0/0	Prometheus Web UI
9100	TCP	0.0.0.0/0	Node Exporter metrics
3030	TCP	0.0.0.0/0	Grafana Web UI

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroupgroup-id=sg-001482c8b9e400baf

aws [Search] [Alt+S] United States (N. Virginia) WahajDev (597532483198) WahajDev

EC2 > Security Groups > sg-001482c8b9e400baf - launch-wizard-1

EC2

- Dashboard
- AWS Global View
- Events
- ▼ **Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- ▼ **Images**
 - AMIs
 - AMI Catalog
- ▼ **Elastic Block Store**
 - Volumes
 - Snapshots
 - Lifecycle Manager
- ▼ **Network & Security**

Inbound security group rules successfully modified on security group (sg-001482c8b9e400baf | launch-wizard-1)

Details

Security group name launch-wizard-1

Security group ID sg-001482c8b9e400baf

Description launch-wizard-1 created 2026-04-23T06:30:28.042Z

VPC ID vpc-030a5ff03bfeea574

Owner 597532483198

Inbound rules count 7 Permission entries

Outbound rules count 1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (7)

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-059cf27b2a569edb7	IPv4	Custom TCP	TCP	8080	0.0.0.0/0
☐	-	sgr-08cd564d2c8eba8bc	IPv4	Custom TCP	TCP	9090	0.0.0.0/0
☐	-	sgr-097fd70af717bd02f	IPv4	Custom TCP	TCP	9100	0.0.0.0/0
☐	-	sgr-0e00b81ecd3736e7b	IPv4	Custom TCP	TCP	3030	0.0.0.0/0
☐	-	sgr-0023c49ece4f162d0	IPv4	Custom TCP	TCP	3001	0.0.0.0/0
☐	-	sgr-042e65108aa4e9159	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-0cfb34792fe283ad2	IPv4	Custom TCP	TCP	3000	0.0.0.0/0

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Part 1: Install Prometheus

Prometheus is a time-series database that collects and stores metrics. It will scrape your EC2 server and Jenkins.

Step 1.1 - Connect to Your EC2 via PowerShell

Open PowerShell on your Windows machine and connect:

```
# Replace with YOUR .pem key file path and EC2 public IP
ssh -i C:\path\to\your-key.pem ubuntu@100.53.31.202

# Once connected, you should see a prompt like:
# ubuntu@ip-100-53-31-202:~$
```

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\wahaj> ssh -i "C:\Users\wahaj\Downloads\jenkey.pem" ubuntu@172.31.20.9
ssh: connect to host 172.31.20.9 port 22: Connection timed out
PS C:\Users\wahaj> ssh -i "C:\Users\wahaj\Downloads\jenkey.pem" ubuntu@100.53.31.202
The authenticity of host '100.53.31.202 (100.53.31.202)' can't be established.
ED25519 key fingerprint is SHA256:eshAH0R+LWgUR8vVa2MK8cu9+PR9/GdLRHcEuJgS+o4.
This host key is known by the following other names/addresses:
  C:\Users\wahaj/.ssh/known_hosts:20: 34,230,36,174
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '100.53.31.202' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Thu May 7 08:40:29 UTC 2026

```
System load:  0.0           Temperature:   -273.1 C
Usage of /:    70.8% of 12.52GB Processes:      155
Memory usage: 10%          Users logged in: 0
Swap usage:   0%           IPv4 address for enp39s0: 172.31.20.9
```

Expanded Security Maintenance for Applications is not enabled.

67 updates can be applied immediately.
56 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

Last login: Thu Apr 23 06:46:12 2026 from 119.154.13.103

```
ubuntu@ip-172-31-20-9:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 07:59:14 UTC; 46min ago
   Main PID: 568 (java)
   Tasks: 55 (limit: 9278)
   Memory: 599.5M (peak: 609.2M)
```

Step 1.2 - Create Prometheus System User

We create a dedicated system user for Prometheus (best security practice):

```
sudo useradd --no-create-home --shell /bin/false prometheus
```

Step 1.3 - Download Prometheus

Download the latest Prometheus binary:

```
cd /tmp

# Download Prometheus 2.47.0 (stable version)
wget https://github.com/prometheus/prometheus/releases/download/v2.47.0/prometheus-2.47.0.linux-amd64.tar.gz

# Extract the archive
tar -xvf prometheus-2.47.0.linux-amd64.tar.gz

# List to confirm extraction
ls prometheus-2.47.0.linux-amd64/
```

```
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)[140.82.112.3]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/c5680bb7f1-f095-412f-8e17-4e410f346fd?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-05-07T10%3A31Z&rscd=attachment%3Bfilename%3Dprometheus-2.47.0.linux-amd64.tar.gz&rsrcct=application%2Foctet-stream&skoid=96c2d410-5711-43al-aedd-ab1947aa7ab0&sktid=39ba6654-997b-47e9-b12b-9515b8b4de&skt=2026-05-07T09%3A07%3A43Z&ske=2026-05-07T10%3A08%3A13Z&skv=sbskvw-2018-11-09&sig=Dvw9EEPhonAAVzbmmSZHQIqHgmApGhzNQIDBg8oW8I%3Ds&jwt=eYJ0exAiOiJKVlQICjhbgCiojJIUzIIINaj9.eYJCpsMIoiNaXRodWYu29tiYYXXviJoicmvsZWfZS1hc3NldHMuzL2la0ahViNXlmcmVbnRlbmcuY29tIlua2VS1joia2VM5MSIsImV4CiCMGTAC3DE0GNziYMsWiBbmJmiJoxNzc4MTQINDIxLCJyYWRoJjoicmVsZWfZSWFzc2V0CHZhZHvjdcLvbiSiBgcg9iNmVmMud2LuZG93cyBuZXQxPQ.NcjDTeyO8hxgbjeePNQKpf_WLCLP4weYMG5GS0devResponse-content-disposition=attachment%3B%20filename%3Dprometheus-2.47.0.linux-amd64.tar.gz&zresponse-content-type=application%2Foctet-stream [following]
2026-05-07 09:17:01 https://release-assets.githubusercontent.com/github-production-release-asset/6838921/c5680bb7f1-f095-412f-8e17-4e410f346fd?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-05-07T10%3A31Z&rscd=attachment%3Bfilename%3Dprometheus-2.47.0.linux-amd64.tar.gz&rsrcct=application%2Foctet-stream&skoid=96c2d410-5711-43al-aedd-ab1947aa7ab0&sktid=39ba6654-997b-47e9-b12b-9515b8b4de&skt=2026-05-07T09%3A07%3A43Z&ske=2026-05-07T10%3A08%3A13Z&skv=sbskvw-2018-11-09&sig=Dvw9EEPhonAAVzbmmSZHQIqHgmApGhzNQIDBg8oW8I%3Ds&jwt=eYJ0exAiOiJKVlQICjhbgCiojJIUzIIINaj9.eYJCpsMIoiNaXRodWYu29tiYYXXviJoicmvsZWfZS1hc3NldHMuzL2la0ahViNXlmcmVbnRlbmcuY29tIlua2VS1joia2VM5MSIsImV4CiCMGTAC3DE0GNziYMsWiBbmJmiJoxNzc4MTQINDIxLCJyYWRoJjoicmVsZWfZSWFzc2V0CHZhZHvjdcLvbiSiBgcg9iNmVmMud2LuZG93cyBuZXQxPQ.NcjDTeyO8hxgbjeePNQKpf_WLCLP4weYMG5GS0devResponse-content-disposition=attachment%3B%20filename%3Dprometheus-2.47.0.linux-amd64.tar.gz&zresponse-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com ([release-assets.githubusercontent.com])... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com ([release-assets.githubusercontent.com])[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 95721304 (91M) [application/octet-stream]
Saving to: 'prometheus-2.47.0.linux-amd64.tar.gz'

prometheus-2.47.0.linux-amd64.tar.gz          100%[=====]                91.29M   475MB/s    in 0.2s

2026-05-07 09:17:01 (475 MB/s) - 'prometheus-2.47.0.linux-amd64.tar.gz' saved [95721304/95721304]

ubuntu@ip-172-31-28-9:/tmp$ tar -xvf prometheus-2.47.0.linux-amd64.tar.gz
prometheus-2.47.0.linux-amd64/
prometheus-2.47.0.linux-amd64/console Libraries/
prometheus-2.47.0.linux-amd64/console Libraries/menu.lib
prometheus-2.47.0.linux-amd64/console Libraries/prom.lib
prometheus-2.47.0.linux-amd64/prometheus.yml
prometheus-2.47.0.linux-amd64/promtool
prometheus-2.47.0.linux-amd64/LICENSE
prometheus-2.47.0.linux-amd64/prometheus
prometheus-2.47.0.linux-amd64/NOTICE
prometheus-2.47.0.linux-amd64/consoles/
prometheus-2.47.0.linux-amd64/consoles/prometheus-overview.html
prometheus-2.47.0.linux-amd64/consoles/node.html
prometheus-2.47.0.linux-amd64/consoles/node-cpu.html
prometheus-2.47.0.linux-amd64/consoles/index.html.example
prometheus-2.47.0.linux-amd64/consoles/prometheus.html
prometheus-2.47.0.linux-amd64/consoles/node-overview.html
prometheus-2.47.0.linux-amd64/consoles/node-disk.html
ubuntu@ip-172-31-28-9:/tmp$ ls prometheus-2.47.0.linux-amd64/
LICENSE NOTICE console Libraries consoles prometheus prometheus.yml promtool
ubuntu@ip-172-31-28-9:/tmp$
```

Step 1.4 - Move Binaries and Create Directories

```
# Create required directories
sudo mkdir /etc/prometheus
sudo mkdir /var/lib/prometheus

# Move the main binaries
sudo cp /tmp/prometheus-2.47.0.linux-amd64/prometheus /usr/local/bin/
sudo cp /tmp/prometheus-2.47.0.linux-amd64/promtool /usr/local/bin/

# Move console files
sudo cp -r /tmp/prometheus-2.47.0.linux-amd64/consoles /etc/prometheus
sudo cp -r /tmp/prometheus-2.47.0.linux-amd64/console_libraries /etc/prometheus

# Set ownership to prometheus user
sudo chown -R prometheus:prometheus /etc/prometheus
sudo chown -R prometheus:prometheus /var/lib/prometheus
```

```
sudo chown prometheus:prometheus /usr/local/bin/prometheus
sudo chown prometheus:prometheus /usr/local/bin/promtool
```

Step 1.5 - Create Prometheus Configuration File

This is the main config file. It tells Prometheus WHAT to scrape and HOW often:

```
sudo nano /etc/prometheus/prometheus.yml
```

Paste the following content exactly (use Ctrl+Shift+V to paste in nano):

```
global:
  scrape_interval: 15s      # Scrape every 15 seconds
  evaluation_interval: 15s  # Evaluate rules every 15 seconds

scrape_configs:

  # Prometheus scrapes itself
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  # Node Exporter - system metrics (CPU, RAM, Disk)
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['localhost:9100']

  # Jenkins monitoring
  - job_name: 'jenkins'
    metrics_path: '/prometheus'
    static_configs:
      - targets: ['localhost:8080']
```

Save the file: press Ctrl+X, then Y, then Enter

```
# Set ownership of the config file
sudo chown prometheus:prometheus /etc/prometheus/prometheus.yml
```

```
Command Prompt x ubuntu@ip-172-31-20-9:/m % +
```

```
iJ9_eYjPc3Hi0i3naXRodiWtY2vTiiwIyXVXiJoicmVsZfzS1hc3NlIHMuZWZlOAHvIdXNlcmVbnRlnQkUY2vTiiaia2VSijoi2a2VMSisImV4cCI6MTczODE0NzIyYSwibm3mjoxNzc4MTQ1NDIxLdCJwYXRJoicmVsZWFzZWZfc2V0CHZHVH  
jdGls15iB694LMvMclud2luZG93cySuZXQ1fQ_NcJDtt9yo8xhgBjeePNQkPf_WlCLP4eYoMSgs8odeYresponse-content-disposition=attachment%3Bfilename%3Dprometheus-2.47.0.linux-amd64.tar.gz&response-co  
ontent-type=application%2Foctet-stream  
Resolving release-assets.githubusercontent.com [release-assets.githubusercontent.com]... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...  
connecting to release-assets.githubusercontent.com [release-assets.githubusercontent.com][185.199.108.133]:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 95721304 (91M) [application/octet-stream]  
Saving to: 'prometheus-2.47.0.linux-amd64.tar.gz'  
  
prometheus-2.47.0.linux-amd64.tar.gz      100%[=====] 91.29M   475MB/s   in 0.2s  
  
2026-05-07 09:17:01 (4975 MB/s) - 'prometheus-2.47.0.linux-amd64.tar.gz' saved [95721304/95721304]  
  
ubuntu@ip-172-31-20-9:/tmp$ tar -xvf prometheus-2.47.0.linux-amd64.tar.gz  
prometheus-2.47.0.linux-amd64/  
prometheus-2.47.0.linux-amd64/console_libraries/  
prometheus-2.47.0.linux-amd64/console_libraries/menu.lib  
prometheus-2.47.0.linux-amd64/console_libraries/prom.lib  
prometheus-2.47.0.linux-amd64/prometheus.yml  
prometheus-2.47.0.linux-amd64/promtool  
prometheus-2.47.0.linux-amd64/LICENSE  
prometheus-2.47.0.linux-amd64/prometheus  
prometheus-2.47.0.linux-amd64/NOTICE  
prometheus-2.47.0.linux-amd64/CONSOLE  
prometheus-2.47.0.linux-amd64/consoles/prometheus-overview.html  
prometheus-2.47.0.linux-amd64/consoles/node.html  
prometheus-2.47.0.linux-amd64/consoles/node-cpu.html  
prometheus-2.47.0.linux-amd64/consoles/index.html.example  
prometheus-2.47.0.linux-amd64/consoles/prometheus.html  
prometheus-2.47.0.linux-amd64/consoles/node-overview.html  
prometheus-2.47.0.linux-amd64/consoles/node-disk.html  
ubuntu@ip-172-31-20-9:/tmp$ ls prometheus-2.47.0.linux-amd64/  
LICENSE NOTICE console_libraries consoles prometheus prometheus.yml promtool  
ubuntu@ip-172-31-20-9:/tmp$ sudo mkdir /etc/prometheus  
ubuntu@ip-172-31-20-9:/tmp$ sudo mkdir /var/lib/prometheus  
ubuntu@ip-172-31-20-9:/tmp$ sudo cp /tmp/prometheus-2.47.0.linux-amd64/prometheus /usr/local/bin/  
ubuntu@ip-172-31-20-9:/tmp$ sudo cp /tmp/prometheus-2.47.0.linux-amd64/promtool /usr/local/bin/  
ubuntu@ip-172-31-20-9:/tmp$ sudo cp -r /tmp/prometheus-2.47.0.linux-amd64/consoles /etc/prometheus  
ubuntu@ip-172-31-20-9:/tmp$ sudo cp -r /tmp/prometheus-2.47.0.linux-amd64/console_libraries /etc/prometheus  
ubuntu@ip-172-31-20-9:/tmp$ sudo chown -R prometheus:prometheus /var/lib/prometheus  
ubuntu@ip-172-31-20-9:/tmp$ sudo chown prometheus:prometheus /usr/local/bin/prometheus  
  
ubuntu@ip-172-31-20-9:/tmp$ sudo chown prometheus:prometheus /etc/prometheus/prometheus.yml  
ubuntu@ip-172-31-20-9:/tmp$
```

Step 1.6 - Create Prometheus Systemd Service

This makes Prometheus run automatically and restart if it crashes:

```
sudo nano /etc/systemd/system/prometheus.service
```

Paste this content:

```
[Unit]
Description=Prometheus Monitoring
Wants=network-online.target
After=network-online.target

[Service]
User=prometheus
Group=prometheus
Type=simple
ExecStart=/usr/local/bin/prometheus \
--config.file=/etc/prometheus/prometheus.yml \
--storage.tsdb.path=/var/lib/prometheus/ \
--web.console.templates=/etc/prometheus/consoles \
--web.console.libraries=/etc/prometheus/console_libraries \
--web.listen-address=0.0.0.0:9090

[Install]
WantedBy=multi-user.target
```

Save: Ctrl+X, then Y, then Enter

Step 1.7 - Start and Enable Prometheus

```
# Reload systemd to recognize new service
sudo systemctl daemon-reload

# Start Prometheus
sudo systemctl start prometheus

# Enable it to start on server reboot
sudo systemctl enable prometheus

# Check status - should show 'active (running)'
sudo systemctl status prometheus
```

```
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl daemon-reload
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl start prometheus
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl status prometheus
● prometheus.service - Prometheus Monitoring
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 09:29:12 UTC; 39s ago
     Main PID: 5964 (prometheus)
        Tasks: 7 (limit: 9278)
       Memory: 24.1M (peak: 26.1M)
          CPU: 68ms
     CGroup: /system.slice/prometheus.service
             └─5964 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus/ --web.console.templates=/etc/prometheus/consoles --web

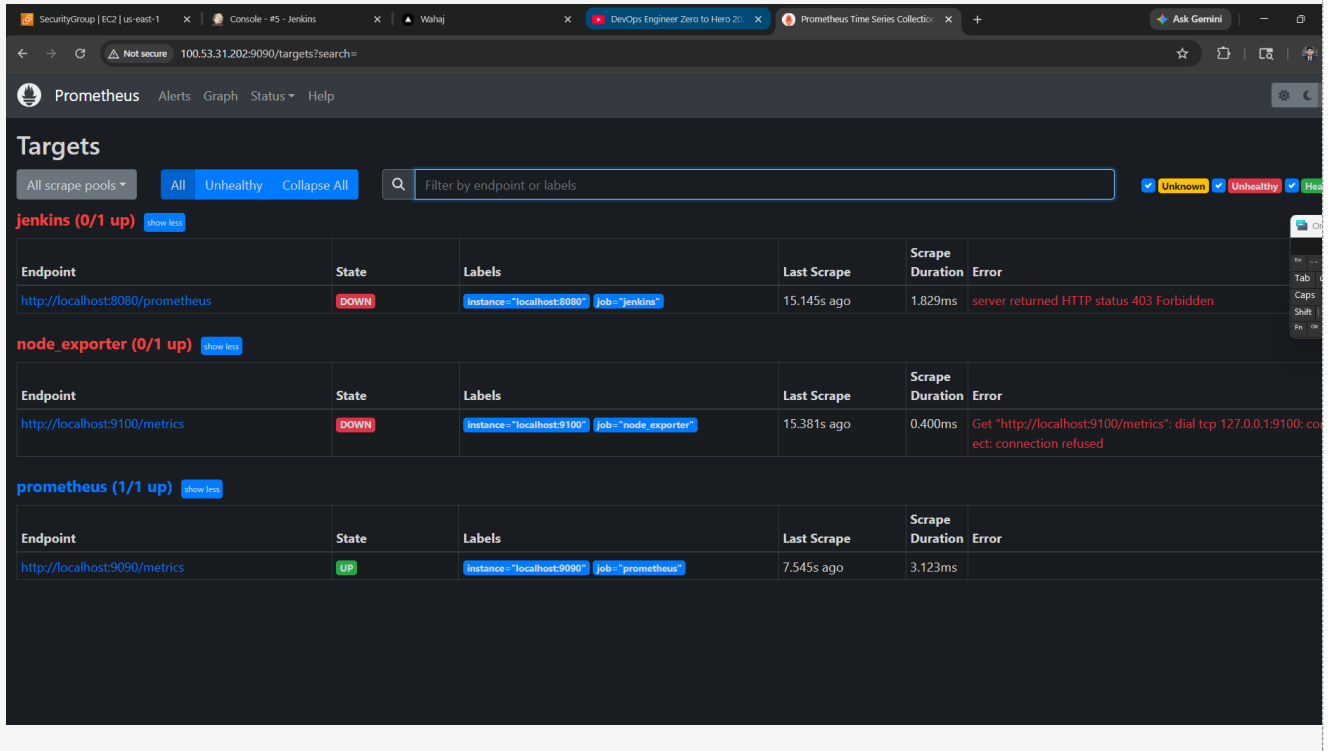
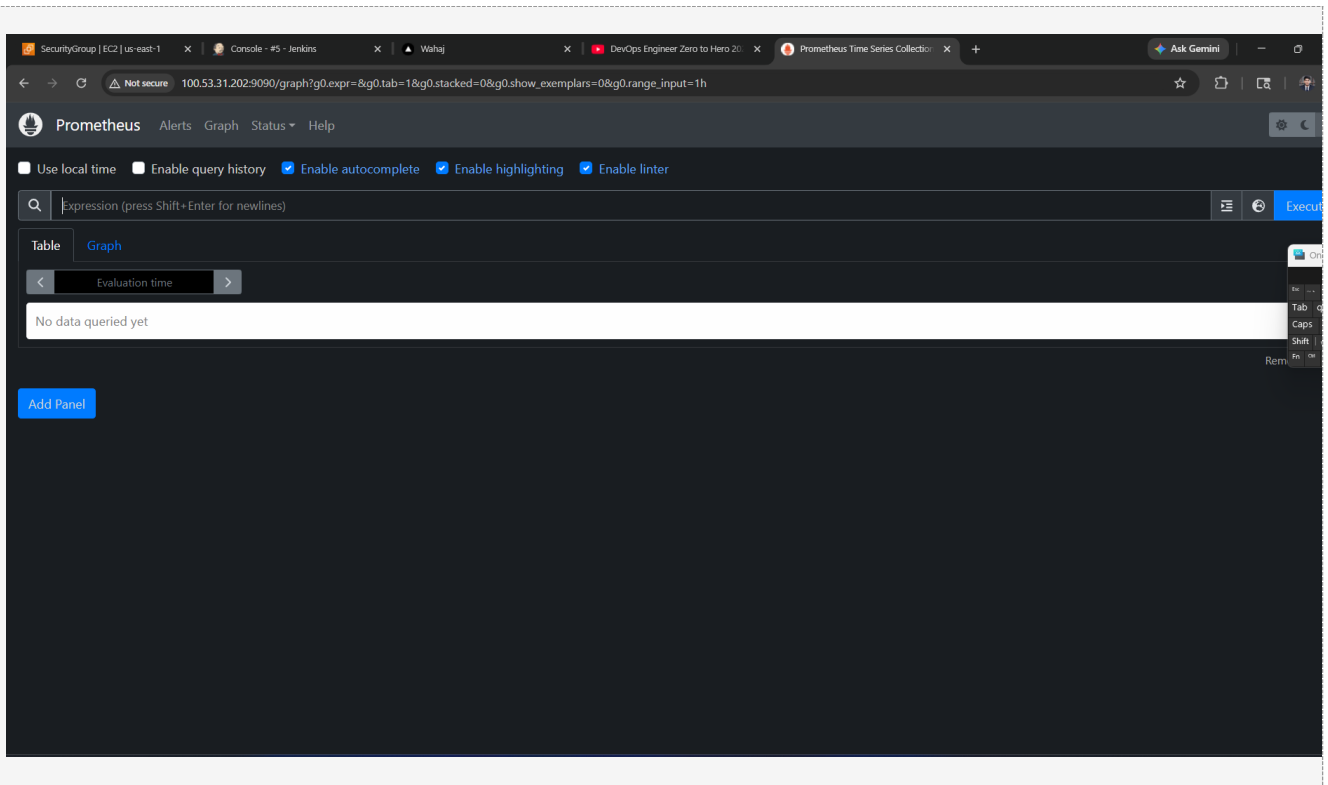
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.531Z caller=head.go:681 level=info component=tsdb msg="On-disk memory mappable chunks replay completed" duration=2
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.531Z caller=head.go:689 level=info component=tsdb msg="Replaying WAL, this may take a while"
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.531Z caller=head.go:760 level=info component=tsdb msg="WAL segment loaded" segment=0 maxSegment=0
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.531Z caller=head.go:797 level=info component=tsdb msg="WAL replay completed" checkpoint_replay_duration=18.601µs v
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.533Z caller=main.go:1045 level=info fs_type=EXT4_SUPER_MAGIC
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.533Z caller=main.go:1048 level=info msg="TSDB started"
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.533Z caller=main.go:1229 level=info msg="Loading configuration file" filename=/etc/prometheus/prometheus.yml
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.535Z caller=main.go:1266 level=info msg="Completed loading of configuration file" filename=/etc/prometheus/prometh
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.535Z caller=main.go:1009 level=info msg="Server is ready to receive web requests."
May 07 09:29:12 ip-172-31-20-9 prometheus[5964]: ts=2026-05-07T09:29:12.535Z caller=manager.go:1009 level=info component="rule manager" msg="Starting rule manager..."
lines 1-20/20 (END)
```

Step 1.8 - Verify Prometheus is Running

Open your browser and go to:

<http://100.53.31.202:9090>

You should see the Prometheus web interface. Click on Status > Targets to verify scrape targets.



Part 2: Install Node Exporter

Node Exporter is a small agent that exposes your EC2 server's system metrics (CPU, Memory, Disk, Network) to Prometheus.

Step 2.1 - Download and Install Node Exporter

```
# Create system user for Node Exporter
sudo useradd --no-create-home --shell /bin/false node_exporter

# Download Node Exporter
cd /tmp
wget
https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz

# Extract
tar -xvf node_exporter-1.6.1.linux-amd64.tar.gz

# Move binary
sudo cp /tmp/node_exporter-1.6.1.linux-amd64/node_exporter /usr/local/bin/

# Set ownership
sudo chown node_exporter:node_exporter /usr/local/bin/node_exporter
```

Step 2.2 - Create Node Exporter Service

```
sudo nano /etc/systemd/system/node_exporter.service
```

Paste this content:

```
[Unit]
Description=Node Exporter
Wants=network-online.target
After=network-online.target

[Service]
User=node_exporter
Group=node_exporter
Type=simple
ExecStart=/usr/local/bin/node_exporter

[Install]
WantedBy=multi-user.target
```

Step 2.3 - Start Node Exporter

```
sudo systemctl daemon-reload
sudo systemctl start node_exporter
```

```
sudo systemctl enable node_exporter
sudo systemctl status node_exporter
```

```

Command Prompt      ubuntu@ip-172-31-20-9: /$  +  ubuntu@ip-172-31-20-9: /$  +  ubuntu@ip-172-31-20-9: /$  +
HjVzHVjdGLv15bIG9iLmVmcUud2LuzG93cySuZX0qfQ.COHPdRmNRZ32vuxE4hBpivXboXlMpkQyEDZ3fZc&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.109.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10368103 (9.9M) [application/octet-stream]
Saving to: 'node_exporter-1.6.1.linux-amd64.tar.gz.1'

node_exporter-1.6.1.linux-amd64.tar.gz.1      100%[=====] 9.89M --.-KB/s  in 0.02s

2026-05-07 09:35:34 (459 MB/s) - 'node_exporter-1.6.1.linux-amd64.tar.gz.1' saved [10368103/10368103]

ubuntu@ip-172-31-20-9:/tmp$ tar -xvf node_exporter-1.6.1.linux-amd64.tar.gz
node_exporter-1.6.1.linux-amd64/
node_exporter-1.6.1.linux-amd64/NOTICE
node_exporter-1.6.1.linux-amd64/node_exporter
node_exporter-1.6.1.linux-amd64/LICENSE
ubuntu@ip-172-31-20-9:/tmp$ sudo cp /tmp/node_exporter-1.6.1.linux-amd64/node_exporter /usr/local/bin/
ubuntu@ip-172-31-20-9:/tmp$ sudo chown node_exporter:node_exporter /usr/local/bin/node_exporter
ubuntu@ip-172-31-20-9:/tmp$ sudo nano /etc/systemd/system/node_exporter.service
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl daemon-reload
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl start node_exporter
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/systemd/system/node_exporter.service.
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 09:37:40 UTC; 19s ago
     Main PID: 6634 (node_exporter)
       Tasks: 5 (limit: 9278)
      Memory: 5.4M (peak: 5.8M)
         CPU: 15ms
       CGroup: /system.slice/node_exporter.service
               └─6634 /usr/local/bin/node_exporter

May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=thermal_zone
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=time
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=timex
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=udp_queues
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=username
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=vmstat
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=yafs
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=node_exporter.go:117 level=info collector=zfs
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=tls_config.go:274 level=info msg="Listening on" address=[::]:9100
May 07 09:37:40 ip-172-31-20-9 node_exporter[6634]: ts=2026-05-07T09:37:40.738Z caller=tls_config.go:277 level=info msg="TLS is disabled." http2=false address=[::]:9100
ubuntu@ip-172-31-20-9:/tmp$

```

Step 2.4 - Verify Node Exporter Metrics

Open browser and visit:

<http://100.53.31.202:9100/metrics>

You should see a long list of system metrics in plain text. This is what Prometheus will scrape.

SecurityGroup [EC2] us-east-1 x Console - #5 - Jenkins x Wahaj x DevOps Engineer Zero to Hero x Prometheus Time Series Coller x 100.53.31.202:9100/metrics x + Ask Gemini

← → ↻ 🔒 Not secure 100.53.31.202:9100/metrics ☆ 📄 🔍 🛠

```
# HELP node_vmstat_pgfault /proc/vmstat information field pgfault.
# TYPE node_vmstat_pgfault untyped
node_vmstat_pgfault 1.752957e06
# HELP node_vmstat_pgmafault /proc/vmstat information field pgmafault.
# TYPE node_vmstat_pgmafault untyped
node_vmstat_pgmafault 7007
# HELP node_vmstat_pgpgin /proc/vmstat information field pgpgin.
# TYPE node_vmstat_pgpgin untyped
node_vmstat_pgpgin 919226
# HELP node_vmstat_pgpgout /proc/vmstat information field pgpgout.
# TYPE node_vmstat_pgpgout untyped
node_vmstat_pgpgout 1.868342e+06
# HELP node_vmstat_pswpin /proc/vmstat information field pswpin.
# TYPE node_vmstat_pswpin untyped
node_vmstat_pswpin 0
# HELP node_vmstat_pswpout /proc/vmstat information field pswpout.
# TYPE node_vmstat_pswpout untyped
node_vmstat_pswpout 0
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 0.83
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 524288
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 10
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 1.8669568e+07
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.77814666022e+09
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 7.4286694e+08
# HELP process_virtual_memory_max_bytes Maximum amount of virtual memory available in bytes.
# TYPE process_virtual_memory_max_bytes gauge
process_virtual_memory_max_bytes 1.8446744073709552e+19
# HELP promhttp_metric_handler_errors_total Total number of internal errors encountered by the promhttp metric handler.
# TYPE promhttp_metric_handler_errors_total counter
promhttp_metric_handler_errors_total{cause="encoding"} 0
promhttp_metric_handler_errors_total{cause="gathering"} 0
# HELP promhttp_metric_handler_requests_in_flight Current number of scrapes being served.
# TYPE promhttp_metric_handler_requests_in_flight gauge
promhttp_metric_handler_requests_in_flight 1
# HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status code.
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 3
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
```

Part 3: Install Grafana

Grafana is the visualization tool. It connects to Prometheus and creates beautiful dashboards. We install it on the SAME EC2 instance but on port 3030 (since 3000 is occupied by your Docker web app).

Step 3.1 - Add Grafana Repository and Install

```
# Install required dependencies
sudo apt-get install -y apt-transport-https software-properties-common wget

# Add Grafana GPG key
sudo mkdir -p /etc/apt/keyrings
wget -q -O - https://apt.grafana.com/gpg.key | gpg --dearmor | sudo tee
/etc/apt/keyrings/grafana.gpg > /dev/null

# Add Grafana repository
echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable
main" | sudo tee /etc/apt/sources.list.d/grafana.list

# Update package list
sudo apt-get update

# Install Grafana
sudo apt-get install -y grafana
```

```
node-optionator node-osenv node-p-cancelable node-p-limit node-p-locate node-p-map node-pascalcase node-path-dirname node-path-exists node-path-is-absolute node-path-is-inside
node-path-type node-pify node-plg-dir node-postcss-value-parser node-prelude-ls node-process-nexttick-args node-promise-inflight node-promise-retry node-promzard node-prr node-pump
node-punycode node-quick-lru node-randombytes node-read node-readable-stream node-rechoir node-regenerator-runtime node-regenerator-transform node-regexp node-regjsen
node-repeat-string node-require-directory node-require-from-string node-resolve node-resolve-cwd node-resolve-from node-restore-cursor node-resumer node-retry node-run-queue
node-safe-buffer node-serialize-javascript node-set-blocking node-set-immediate-shim node-shebang-command node-shebang-regex node-shell-quote node-signal-exit node-slash
node-slice-ansi node-source-list-map node-source-map node-spdx-correct node-spdx-exceptions node-spdx-expression-parse node-spdx-license-ids node-sprintf-js node-sri node-stack-utls
node-string-decoder node-strip-bom node-supports-color node-tapable node-terser node-text-table node-through node-time-stamp node-to-fast-properties node-tslib node-type-check
node-typedarray node-typedarray-to-buffer node-undici node-unicode-canonical-property-names-ecmascript node-unicode-match-property-value-ecmascript
node-unicode-property-aliases-ecmascript node-unset-value node-uri-js node-util-deprecate node-uuid node-v8flags node-validate-npm-package-license node-width.js node-webpack-sources
node-wordwrap node-wrap node-write-file-atomic node-xtd node-y18n node-yallist node-yaml x11-xserver-utils xdg-utils zutty

Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  grafana
0 upgraded, 1 newly installed, 0 to remove and 84 not upgraded.
Need to get 274 MB of archives.
After this operation, 1028 MB of additional disk space will be used.
Get:1 https://apt.grafana.com stable/main amd64 grafana amd64 13.0.1 [274 MB]
Fetched 274 MB in 4s (63.3 MB/s)
Selecting previously unselected package grafana.
(Reading database ... 11561 files and directories currently installed.)
Preparing to unpack .../grafana_13.0.1_amd64.deb ...
Unpacking grafana (13.0.1) ...
Setting up grafana (13.0.1) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user 'grafana' (UID 112) ...
info: Adding new user 'grafana' (UID 112) with group 'grafana' ...
info: Not creating home directory '/usr/share/grafana'.
#### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
  sudo /bin/systemctl daemon-reload
  sudo /bin/systemctl enable grafana-server
#### You can start grafana-server by executing
  sudo /bin/systemctl start grafana-server
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-9:/tmp$ |
```

Step 3.2 - Change Grafana Port to 3030

Since your web app is on port 3000, we change Grafana to use port 3030:

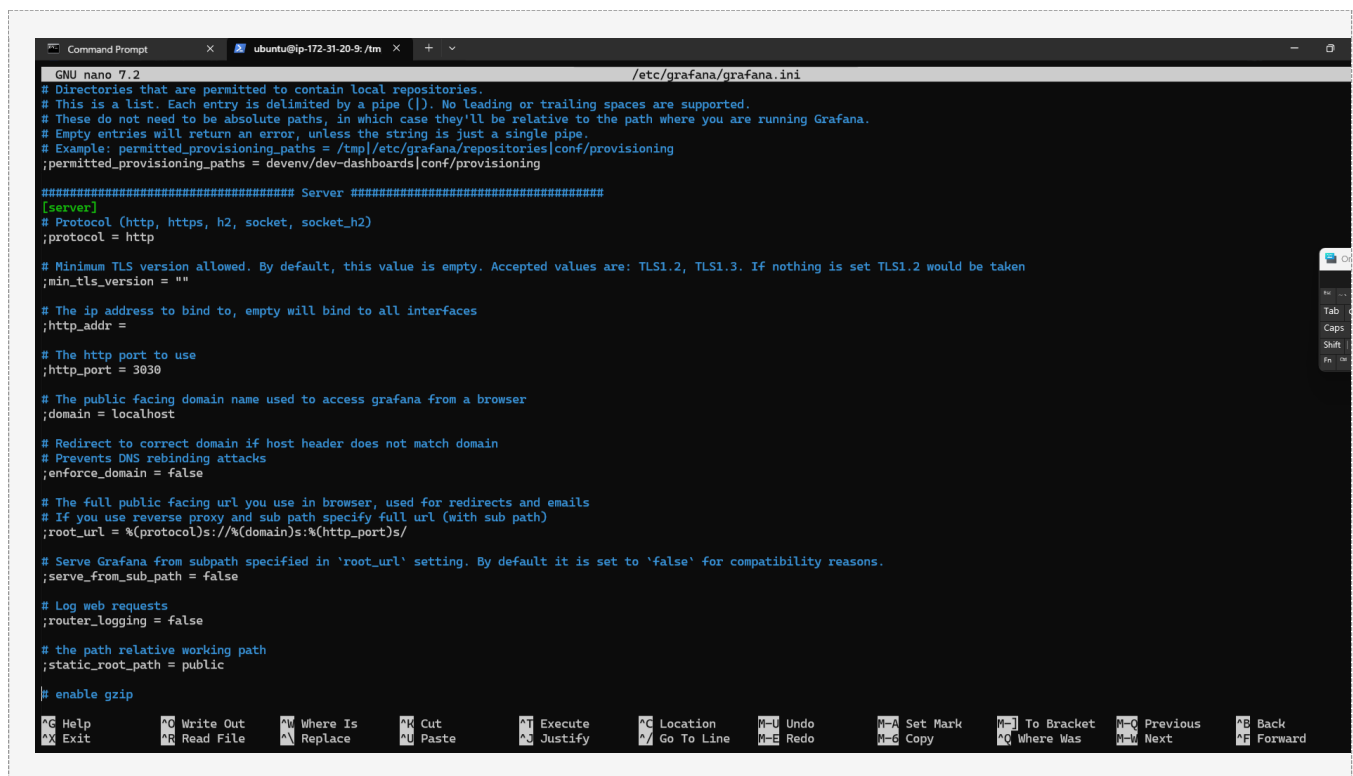
```
sudo nano /etc/grafana/grafana.ini
```

Find the [server] section (press Ctrl+W to search, type 'http_port'). Change the port:

```
[server]
# The IP address to bind to, empty will bind to all interfaces
http_addr =

# The http port to use
http_port = 3030
```

Save: Ctrl+X, then Y, then Enter



```
GNU nano 7.2 /etc/grafana/grafana.ini
# Directories that are permitted to contain local repositories.
# This is a list. Each entry is delimited by a pipe (|). No leading or trailing spaces are supported.
# These do not need to be absolute paths, in which case they'll be relative to the path where you are running Grafana.
# Empty entries will return an error, unless the string is just a single pipe.
# Example: permitted_provisioning_paths = /tmp/etc/grafana/repositories/conf/provisioning
;permitted_provisioning_paths = devenv/dev-dashboards/conf/provisioning

##### Server #####
[server]
# Protocol (http, https, h2, socket, socket_h2)
;protocol = http

# Minimum TLS version allowed. By default, this value is empty. Accepted values are: TLS1.2, TLS1.3. If nothing is set TLS1.2 would be taken
;min_tls_version = ""

# The ip address to bind to, empty will bind to all interfaces
;http_addr =

# The http port to use
;http_port = 3030

# The public facing domain name used to access grafana from a browser
;domain = localhost

# Redirect to correct domain if host header does not match domain
# Prevents DNS rebinding attacks
;enforce_domain = false

# The full public facing url you use in browser, used for redirects and emails
# If you use reverse proxy and sub path specify full url (with sub path)
;root_url = %(protocol)s://%(domain)s:%(http_port)s/

# Serve Grafana from subpath specified in 'root_url' setting. By default it is set to 'false' for compatibility reasons.
;serve_from_sub_path = false

# Log web requests
;router_logging = false

# the path relative working path
;static_root_path = public

# enable gzip

Ctrl+H Help      Ctrl+W Write Out  Ctrl+M Where Is   Ctrl+U Cut         Ctrl+J Execute    Ctrl+L Location   Ctrl+_ Undo         Ctrl+V Set Mark   Ctrl+] To Bracket  Ctrl+Q Previous  Ctrl+B Back
Ctrl+X Exit      Ctrl+R Read File  Ctrl+A Replace    Ctrl+P Paste      Ctrl+K Justify    Ctrl+G Go To Line  Ctrl+Z Redo       Ctrl+C Copy      Ctrl+[ Where Was  Ctrl+V Next     Ctrl+F Forward
```

Step 3.3 - Start and Enable Grafana

```
# Start Grafana service
sudo systemctl start grafana-server

# Enable on boot
sudo systemctl enable grafana-server

# Check status
sudo systemctl status grafana-server
```

```
Command Prompt x ubuntu@ip-172-31-20-9: /tmp x + v
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl restart grafana-server
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl status grafana-server
* grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Thu 2026-05-07 09:49:59 UTC; 7s ago
   Duration: 1.255s
   Docs: http://docs.grafana.org
   Process: 8586 ExecStart=/usr/share/grafana/bin/grafana server --config=${CONF_FILE} --pidfile=${PID_FILE_DIR}/grafana-server.pid --packaging=deb cfg:default.paths.logs=${LOG_DIR} cfg:default.paths.logs=${LOG_DIR}
   Main PID: 8586 (code=exited, status=1/FAILURE)
   CPU: 1.514s

May 07 09:49:59 ip-172-31-20-9 systemd[1]: grafana-server.service: Scheduled restart job, restart counter is at 5.
May 07 09:49:59 ip-172-31-20-9 systemd[1]: grafana-server.service: Start request repeated too quickly.
May 07 09:49:59 ip-172-31-20-9 systemd[1]: grafana-server.service: Failed with result 'exit-code'.
May 07 09:49:59 ip-172-31-20-9 systemd[1]: Failed to start grafana-server.service - Grafana instance.
May 07 09:49:59 ip-172-31-20-9 systemd[1]: grafana-server.service: Consumed 1.514s CPU time.

ubuntu@ip-172-31-20-9:/tmp$ sudo sed -i '/^[server\]/a http_port = 3030' /etc/grafana/grafana.ini
ubuntu@ip-172-31-20-9:/tmp$ sudo grep "http_port" /etc/grafana/grafana.ini
http_port = 3030
;http_port = 3030
;root_url = %(protocol)s://%(domain)s:%(http_port)s/
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl restart grafana-server
ubuntu@ip-172-31-20-9:/tmp$ sudo systemctl status grafana-server
* grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-07 09:51:39 UTC; 8s ago
   Docs: http://docs.grafana.org
   Main PID: 8657 (grafana)
   Tasks: 14 (limit: 9278)
   Memory: 149.8M (peak: 156.8M)
   CPU: 1.498s
   CGroup: /system.slice/grafana-server.service
           └─8657 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/grafana/grafana-server.pid --packaging=deb cfg:default.paths.logs=/var/log/grafana

May 07 09:51:40 ip-172-31-20-9 grafana[8657]: t=2026-05-07T09:51:40.802957077Z level=info caller=logger.go:214 time=2026-05-07T09:51:40.802943052Z msg="Starting ConnectionController" log
May 07 09:51:40 ip-172-31-20-9 grafana[8657]: t=2026-05-07T09:51:40.804378274Z level=info caller=logger.go:214 time=2026-05-07T09:51:40.80436251Z msg="Starting workers" logger=provisioni
May 07 09:51:40 ip-172-31-20-9 grafana[8657]: t=2026-05-07T09:51:40.804408761Z level=info caller=logger.go:214 time=2026-05-07T09:51:40.804405303Z msg="Started workers" logger=provisioni
May 07 09:51:40 ip-172-31-20-9 grafana[8657]: t=2026-05-07T09:51:40.805377992Z level=info caller=logger.go:214 time=2026-05-07T09:51:40.805365887Z msg="Starting workers" logger=provisioni
May 07 09:51:40 ip-172-31-20-9 grafana[8657]: t=2026-05-07T09:51:40.806269584Z level=info caller=logger.go:214 time=2026-05-07T09:51:40.806260411Z msg="Started workers" logger=provisioni
May 07 09:51:40 ip-172-31-20-9 grafana[8657]: logger=plugin.angular detectorsprovider.dynamic t=2026-05-07T09:51:40.89080534Z level=info msg="Patterns update finished" duration=70.08988ms
May 07 09:51:41 ip-172-31-20-9 grafana[8657]: logger=plugin.backgroundinstaller t=2026-05-07T09:51:41.180621818Z level=info msg="Installing plugin" pluginId=elasticsearch version=
May 07 09:51:41 ip-172-31-20-9 grafana[8657]: logger=plugin.installer t=2026-05-07T09:51:41.328492435Z level=info msg="Updating plugin" pluginId=elasticsearch from=12.5.4 to=12.6.1
May 07 09:51:41 ip-172-31-20-9 grafana[8657]: logger=plugin.elasticsearch t=2026-05-07T09:51:41.330027458Z level=info msg="plugin process exited" plugin=/usr/share/grafana/data/plugins-b
May 07 09:51:41 ip-172-31-20-9 grafana[8657]: logger=plugin.backgroundinstaller t=2026-05-07T09:51:41.330119151Z level=error msg="Failed to install plugin" pluginId=elasticsearch version=
lines 1-21/21 (END)
```

Step 3.4 - Access Grafana Web UI

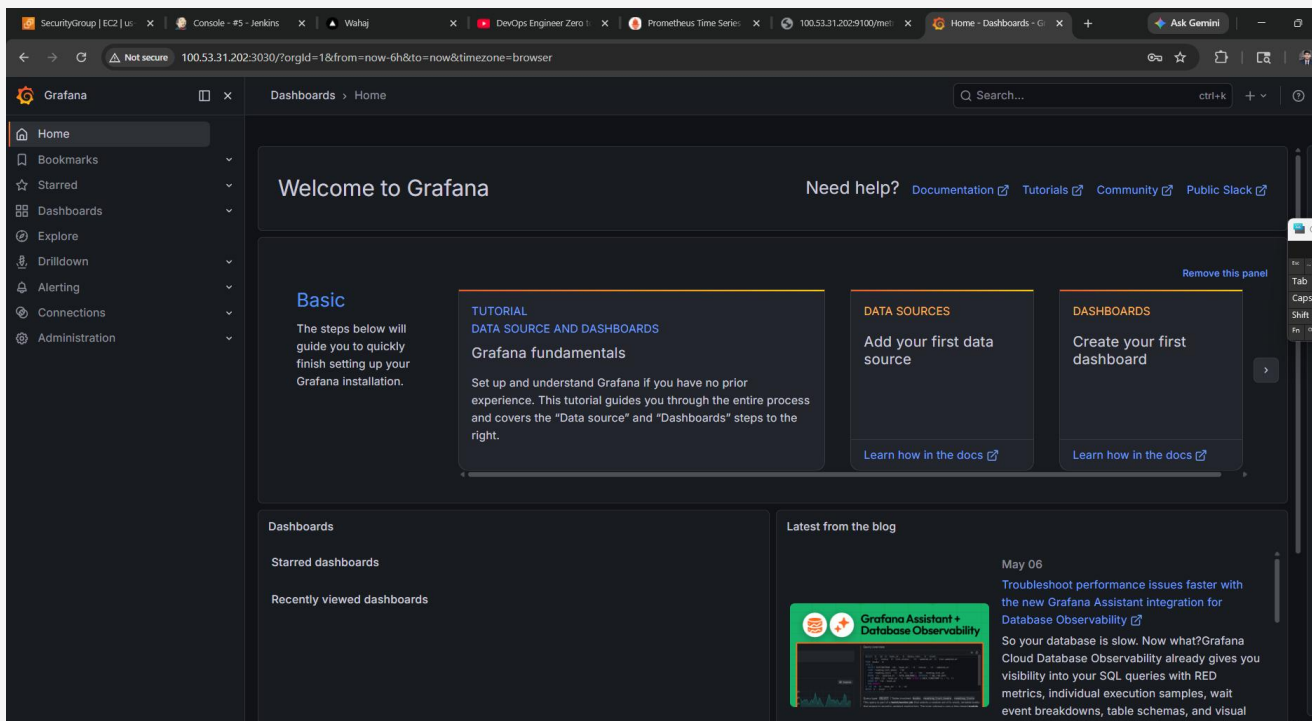
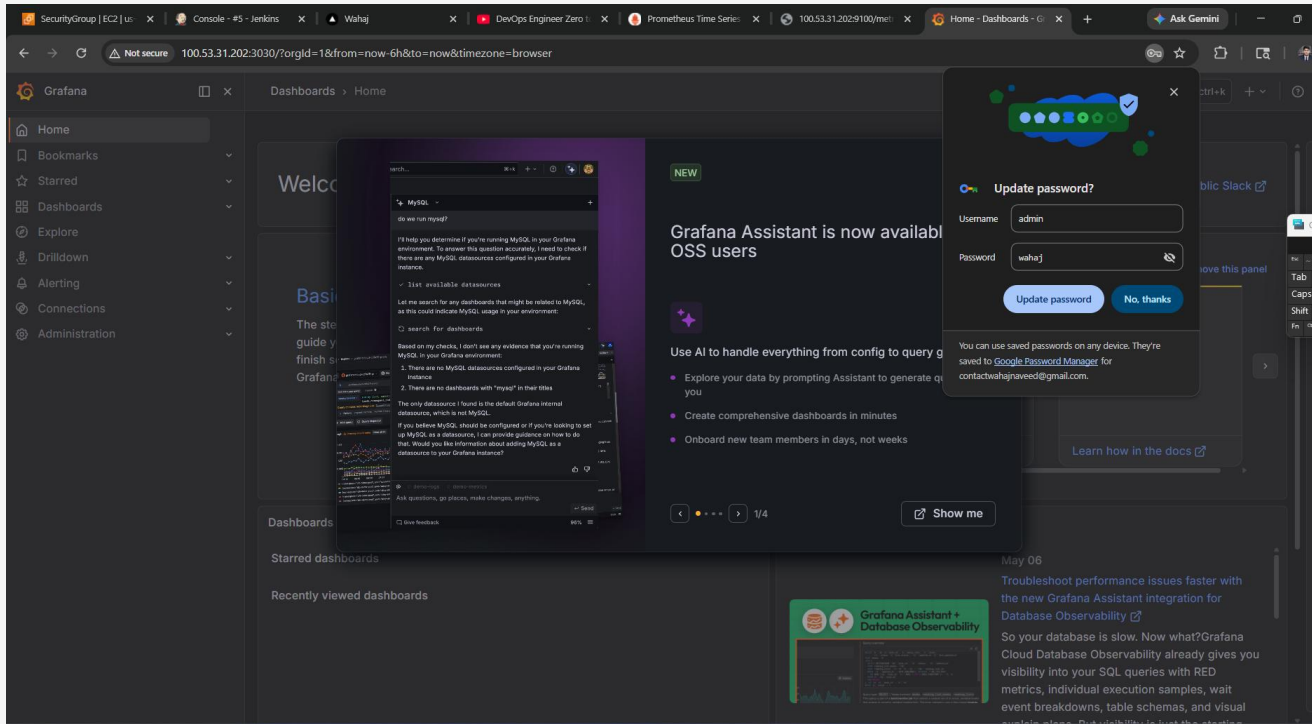
Open your browser and navigate to:

`http://100.53.31.202:3030`

You will see the Grafana login page. Use these default credentials:

Username	admin
Password	admin

After first login, Grafana will ask you to change the password. Set a strong password and remember it!

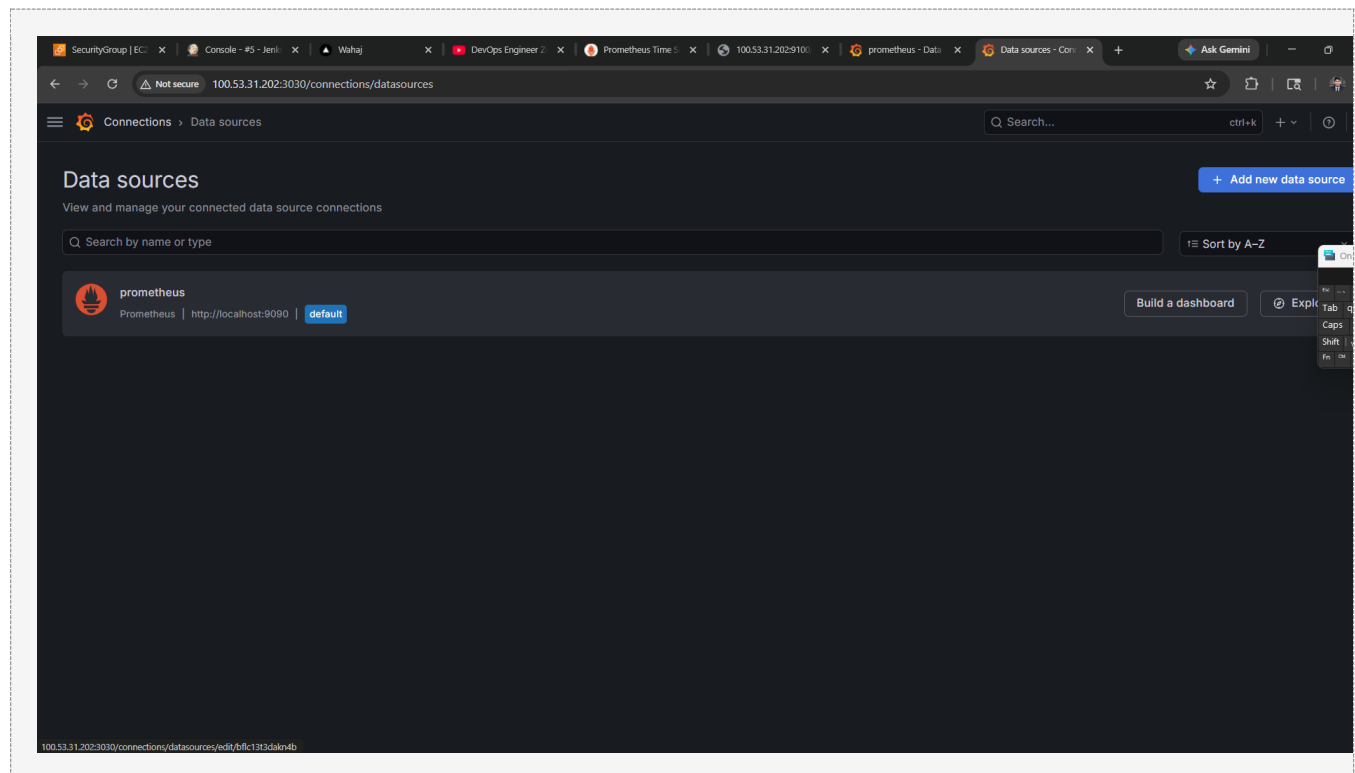


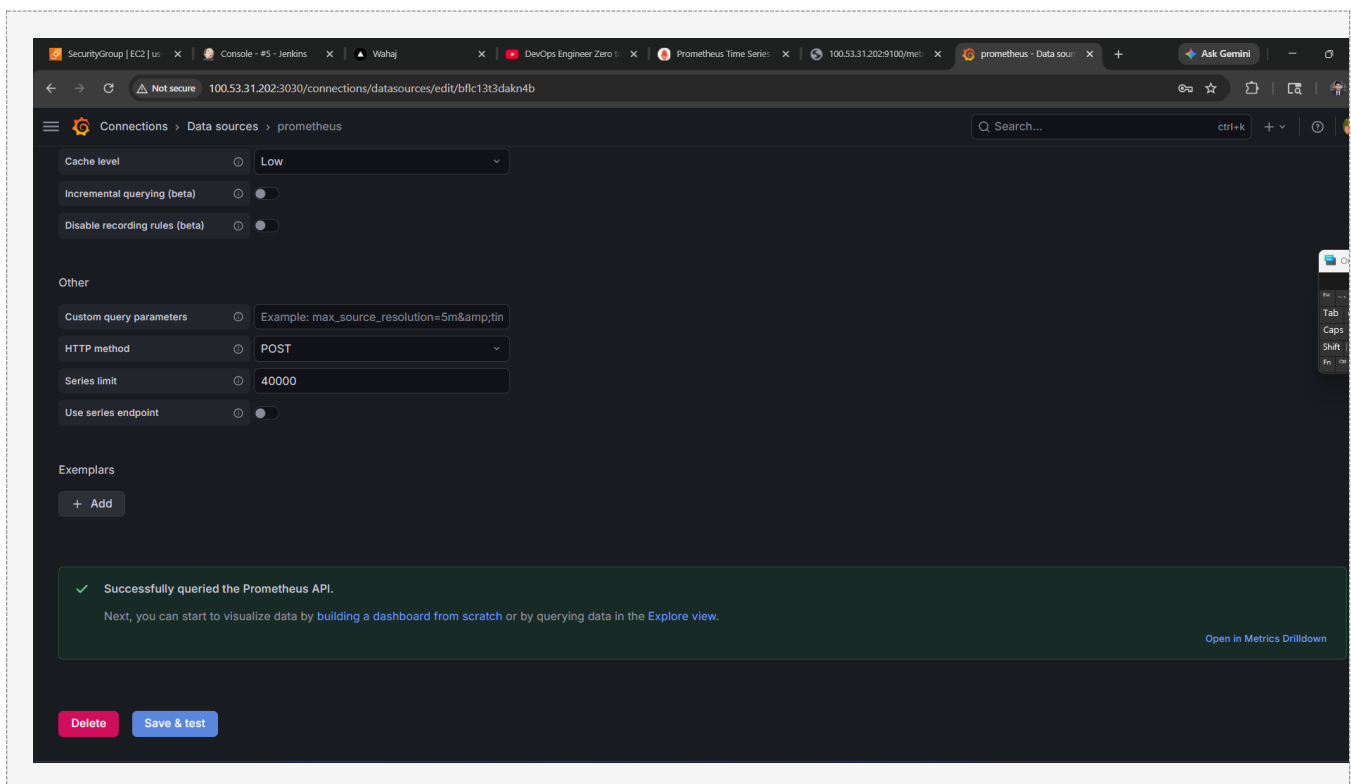
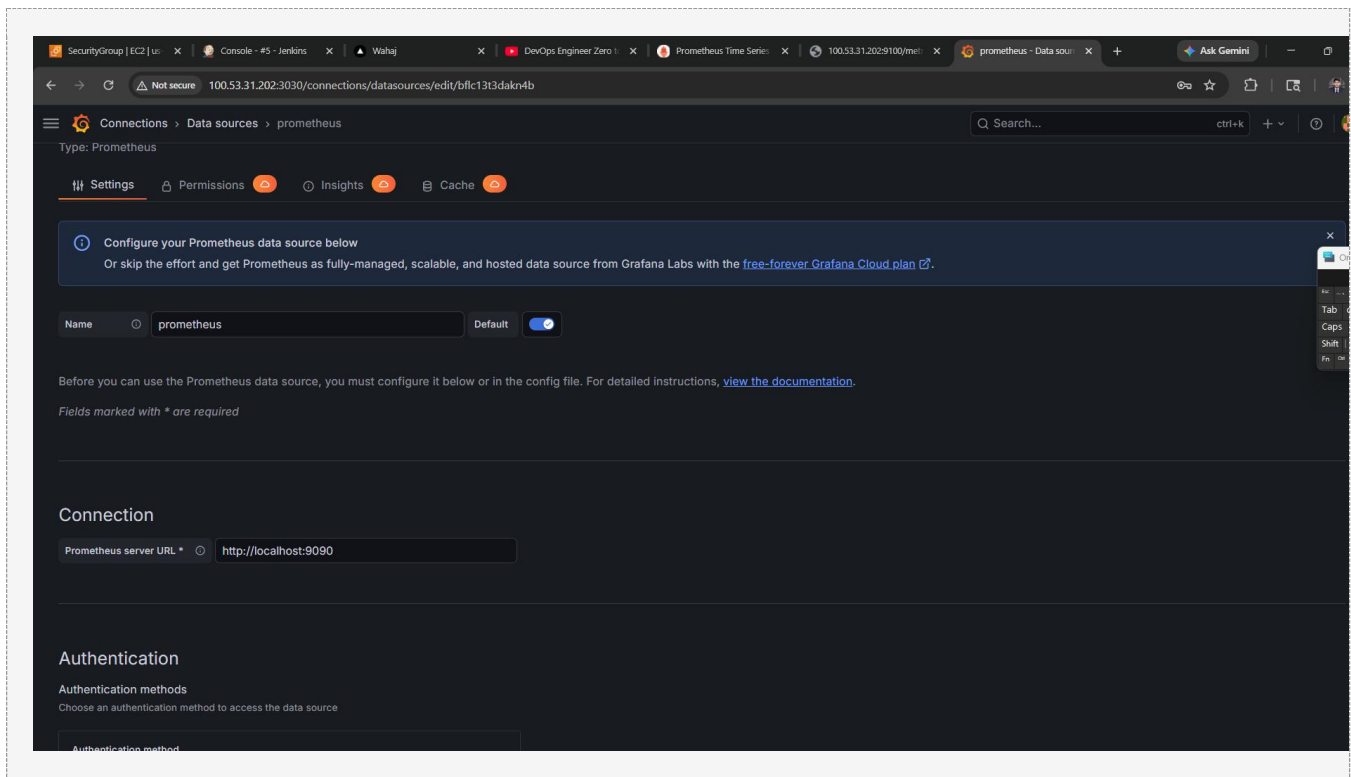
Part 4: Integrate Grafana with Prometheus

Now we connect Grafana to Prometheus so Grafana can read and visualize the metrics data.

Step 4.1 - Add Prometheus as Data Source in Grafana

1	In Grafana, click the gear icon (Configuration) in the left sidebar, then click 'Data Sources'
2	Click the blue 'Add data source' button
3	In the search box, type 'Prometheus' and click on it
4	In the HTTP section, set URL to: <code>http://localhost:9090</code>
5	Leave all other settings as default
6	Scroll down and click 'Save & Test'
7	You should see a green message: 'Data source is working'

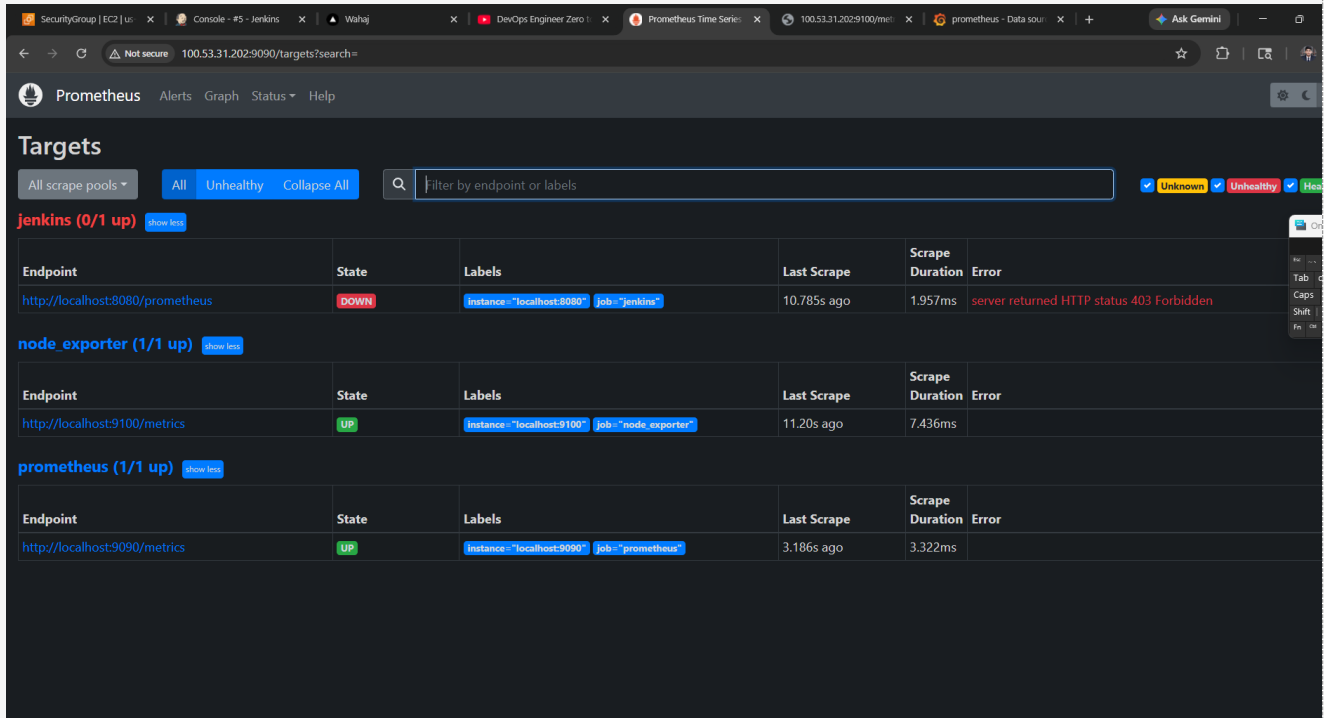




Step 4.2 - Verify Prometheus Targets are UP

Before creating dashboards, confirm that all your scrape targets are healthy:

- 1 Open browser and go to <http://100.53.31.202:9090>
- 2 Click on 'Status' in the top navigation menu
- 3 Click on 'Targets'
- 4 You should see all three jobs (prometheus, node_exporter, jenkins) with State = UP in green



The screenshot shows the Prometheus web interface at the 'Targets' page. The URL is 100.53.31.202:9090/targets?search=. The page displays three scrape pools:

- jenkins (0/1 up)**: The target <http://localhost:8080/prometheus> is DOWN. The error message is 'server returned HTTP status 403 Forbidden'.
- node_exporter (1/1 up)**: The target <http://localhost:9100/metrics> is UP.
- prometheus (1/1 up)**: The target <http://localhost:9090/metrics> is UP.

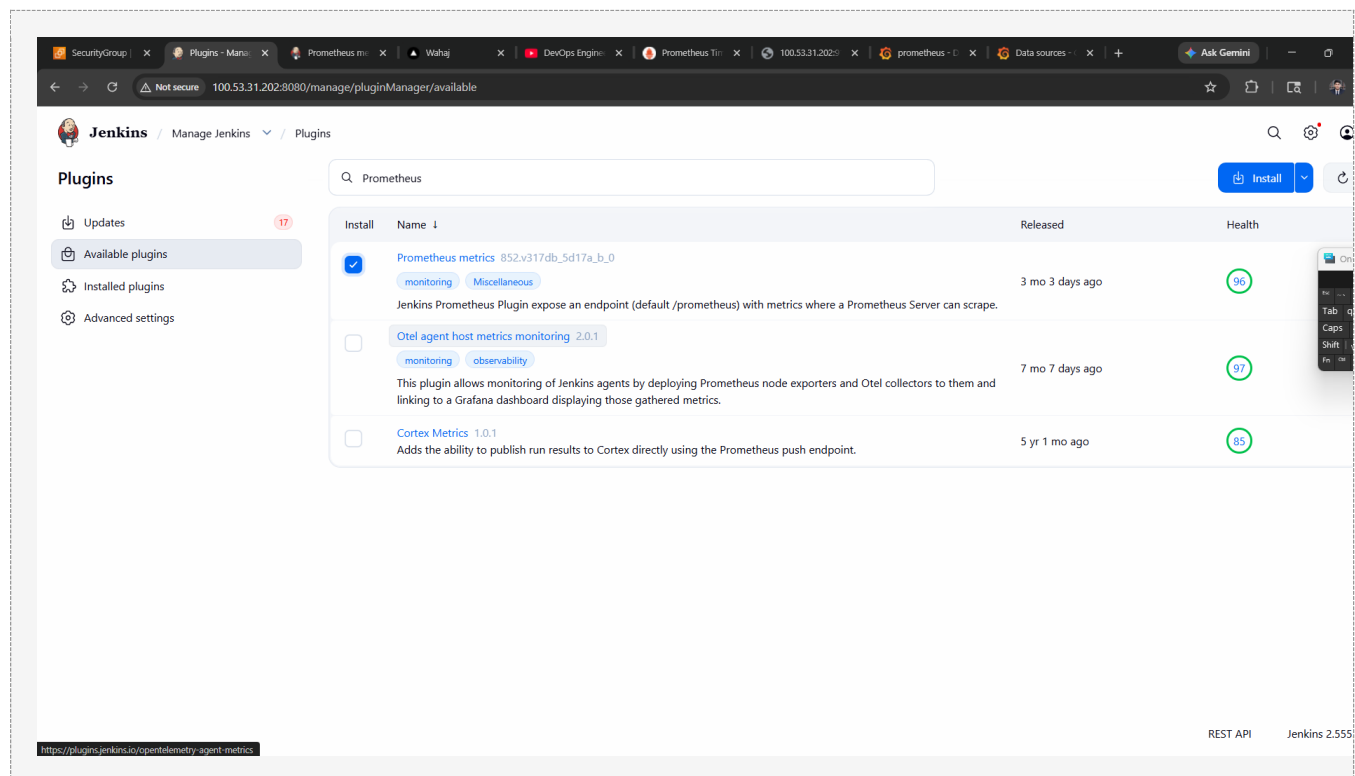
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
jenkins (0/1 up)					
http://localhost:8080/prometheus	DOWN	instance="localhost:8080" job="jenkins"	10.785s ago	1.957ms	server returned HTTP status 403 Forbidden
node_exporter (1/1 up)					
http://localhost:9100/metrics	UP	instance="localhost:9100" job="node_exporter"	11.20s ago	7.436ms	
prometheus (1/1 up)					
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	3.186s ago	3.322ms	

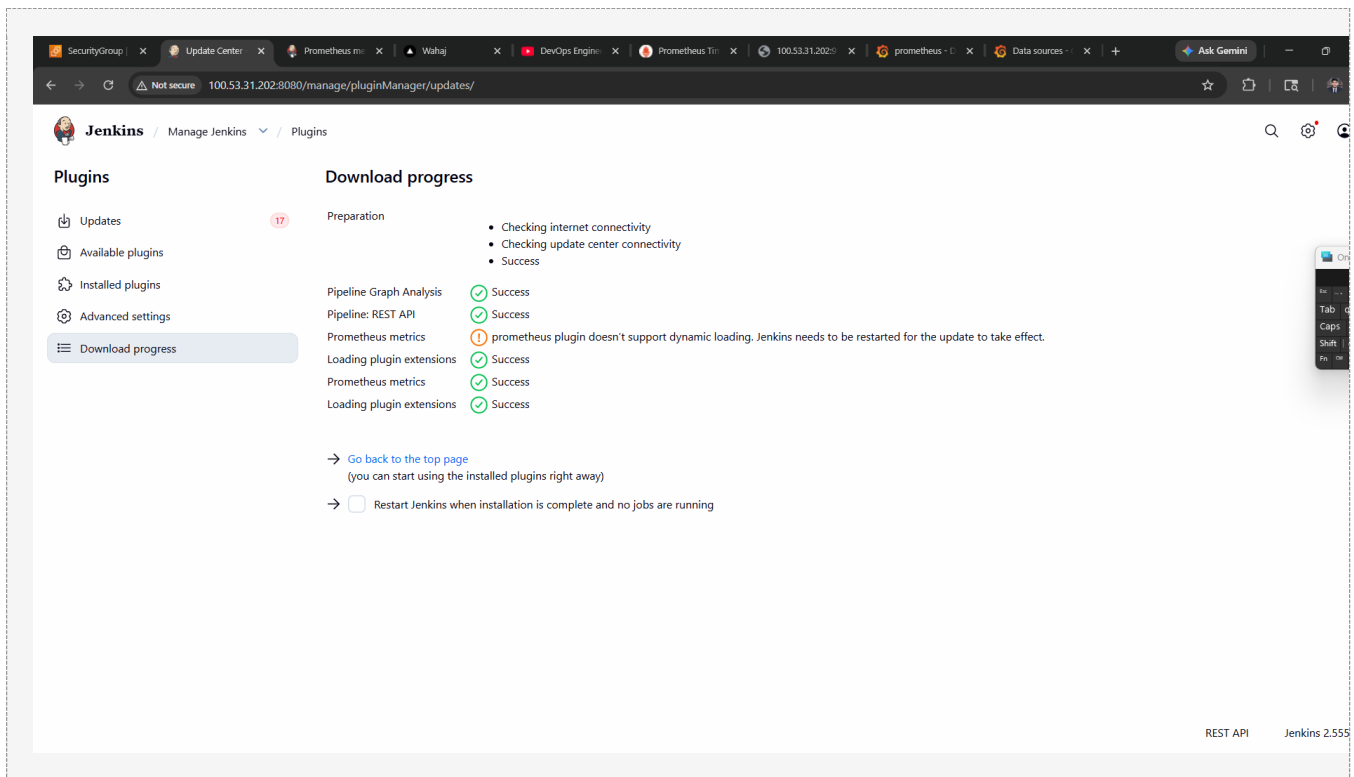
Part 5: Configure Jenkins Prometheus Plugin

To monitor Jenkins jobs and build metrics, install the Prometheus plugin in Jenkins.

Step 5.1 - Install Prometheus Plugin in Jenkins

1	Open Jenkins at http://100.53.31.202:8080/
2	Click 'Manage Jenkins' from the left sidebar
3	Click 'Manage Plugins'
4	Click the 'Available plugins' tab
5	In the search box, type 'Prometheus'
6	Check the box next to 'Prometheus metrics'
7	Click 'Install without restart'
8	Wait for installation to complete





Step 5.2 - Verify Jenkins Metrics Endpoint

After installing the plugin, Jenkins will expose metrics at /prometheus. Verify it works:

`http://100.53.31.202:8080/prometheus`

You should see Jenkins metrics in Prometheus format (lines starting with # HELP, # TYPE, etc.)

SecurityGroup | x | Plugins - Man... x | Wahaj x | DevOps Engine x | Prometheus T... x | 100.53.31.202: x | prometheus - 0 x | Data sources - x | 100.53.31.202: x | + | Ask Gemini | - |

← → ↻ | Not secure | 100.53.31.202:8080/prometheus/ | ☆ | 📄 | 🔍 | 🛠 |

```
# HELP jvm_memory_pool_allocated_bytes_total Total bytes allocated in a given JVM memory pool. Only updated after GC, not continuously.
# TYPE jvm_memory_pool_allocated_bytes_total counter
# HELP jvm_classes_currently_loaded The number of classes that are currently loaded in the JVM
# TYPE jvm_classes_currently_loaded gauge
jvm_classes_currently_loaded 17006.0
# HELP jvm_classes_loaded_total The total number of classes that have been loaded since the JVM has started execution
# TYPE jvm_classes_loaded_total counter
jvm_classes_loaded_total 17006.0
# HELP jvm_classes_unloaded_total The total number of classes that have been unloaded since the JVM has started execution
# TYPE jvm_classes_unloaded_total counter
jvm_classes_unloaded_total 0.0
# HELP jvm_threads_current Current thread count of a JVM
# TYPE jvm_threads_current gauge
jvm_threads_current 69.0
# HELP jvm_threads_daemon Daemon thread count of a JVM
# TYPE jvm_threads_daemon gauge
jvm_threads_daemon 57.0
# HELP jvm_threads_peak Peak thread count of a JVM
# TYPE jvm_threads_peak gauge
jvm_threads_peak 69.0
# HELP jvm_threads_started_total Started thread count of a JVM
# TYPE jvm_threads_started_total counter
jvm_threads_started_total 108.0
# HELP jvm_threads_deadlocked Cycles of JVM-threads that are in deadlock waiting to acquire object monitors or ownable synchronizers
# TYPE jvm_threads_deadlocked gauge
jvm_threads_deadlocked 0.0
# HELP jvm_threads_deadlocked_monitor Cycles of JVM-threads that are in deadlock waiting to acquire object monitors
# TYPE jvm_threads_deadlocked_monitor gauge
jvm_threads_deadlocked_monitor 0.0
# HELP jvm_threads_state Current count of threads by state
# TYPE jvm_threads_state gauge
jvm_threads_state{state="NEW",} 0.0
jvm_threads_state{state="TERMINATED",} 0.0
jvm_threads_state{state="RUNNABLE",} 7.0
jvm_threads_state{state="BLOCKED",} 0.0
jvm_threads_state{state="WAITING",} 15.0
jvm_threads_state{state="TIMED_WAITING",} 47.0
jvm_threads_state{state="UNKNOWN",} 0.0
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 19.21
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.778140490343E9
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 323.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 524288.0
```

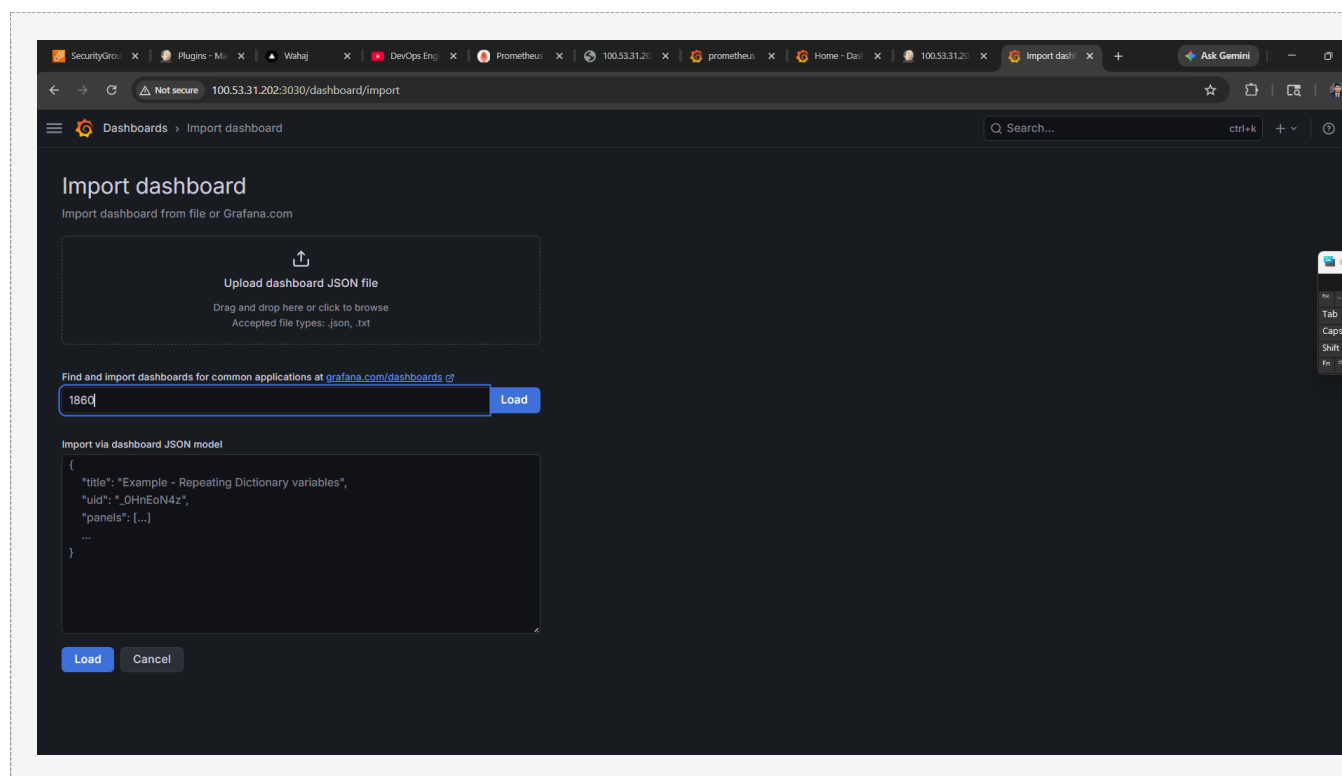
Part 6: Create and Access Dashboards

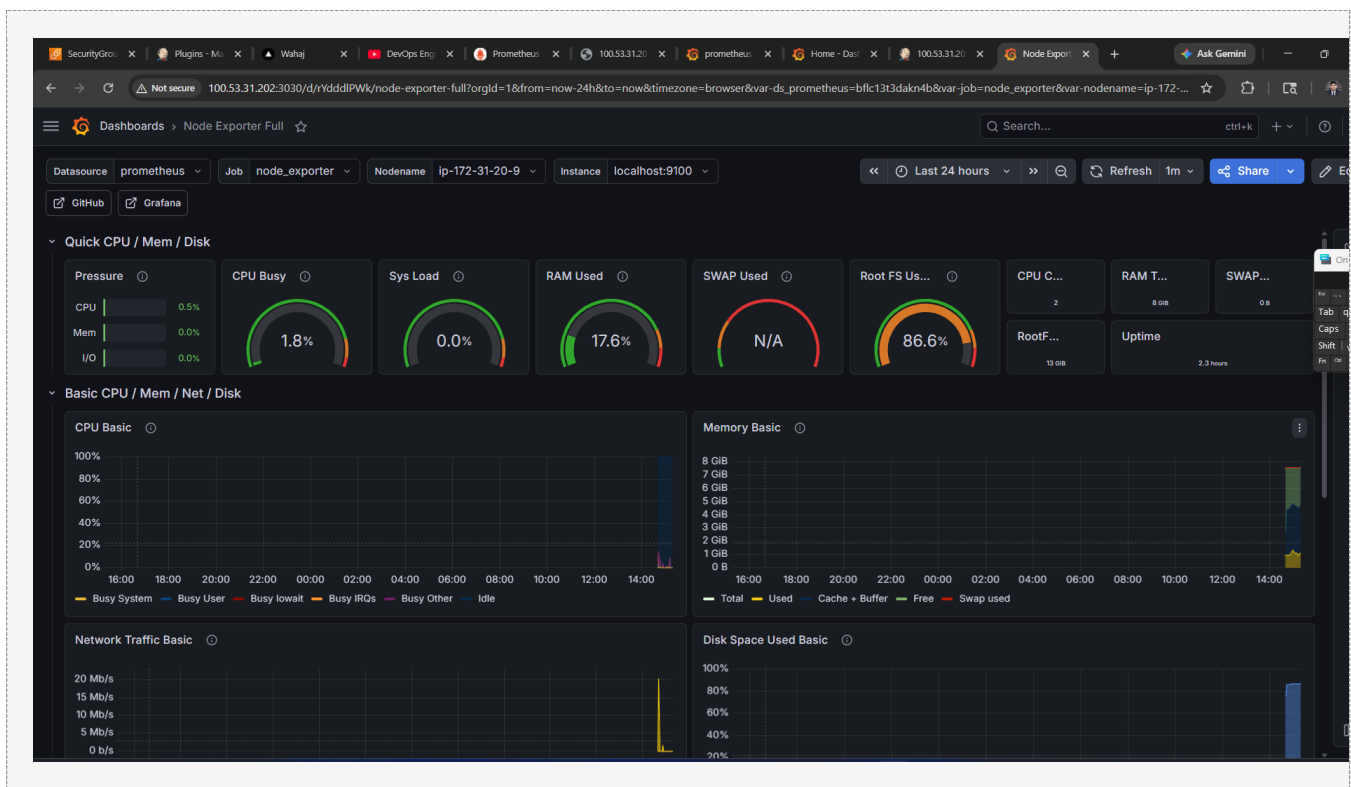
Now we set up pre-built dashboards to monitor your server and Jenkins. Grafana has a library of thousands of community dashboards.

Step 6.1 - Import Node Exporter Dashboard

This gives you a full server monitoring dashboard (CPU, RAM, Disk, Network) in seconds:

1	In Grafana, click the '+' (plus) icon in the left sidebar
2	Click 'Import'
3	In the 'Import via grafana.com' field, type: 1860
4	Click 'Load'
5	In the 'Prometheus' dropdown, select the Prometheus data source you created
6	Click 'Import'
7	Dashboard will load immediately with all server metrics!





Step 6.2 - Import Jenkins Dashboard

Import a pre-built Jenkins monitoring dashboard:

- 1 In Grafana, click '+' icon then 'Import'
- 2 Enter dashboard ID: 9964
- 3 Click 'Load'
- 4 Select your Prometheus data source
- 5 Click 'Import'

SecurityGro...Plugins - M...WahajDevOps EngPrometheus100.53.31.20100.53.31.20100.53.31.20Home - Das100.53.31.20Import dashAsk Gemini

Not secure100.53.31.202:3030/dashboard/import

Search...ctrl+k+ +

Dashboards > Import dashboard

Import dashboard

Import dashboard from file or Grafana.com

Importing dashboard from Grafana.com

Published byharyan

Updated on2023-08-24 14:34:53

Options

NameJenkins: Performance and Health Overview

FolderDashboards

Unique Identifier (UID)
The unique identifier (UID) of a dashboard can be used to uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.
haryan-jenkinsChange uid

DS_PROMETHEUS
Select a prometheus data source
Prometheus Data Source

ImportCancel

